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THE TWO-DIMENSIONAL EIGENVALUE RANGE AND EXTREMAL EIGENVALUE PROBLEMS*

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Abstract. If $a(x)$ and $b(x)$ are measurable functions on [0, 1] satisfying $0<a(x)<b(x)$, let $C(a(x),b(x))$ denote the class of functions $\varphi(x)\in L_1[0,1]$ such that $a(x)\le\varphi(x)\le b(x)$. Given such a $\varphi(x)$, let $\lambda_1(\varphi)$ and $\lambda_2(\varphi)$ denote the first two eigenvalues of the problem $y''+\lambda\varphi(x)y=0$, $y(0)=y(1)=0$. In this paper we characterize the set $(\lambda_1(\varphi),\lambda_2(\varphi))$ as φ varies over $C(a(x),b(x))$. Explicit analytic and numerical results are given when $a(x)$ and $b(x)$ are constant. The connection between this problem and the extremization of functions of eigenvalues is also discussed.

1. Introduction. Consider the eigenvalue problem

$$y''+\lambda\varphi(x)y=0,\qquad y(0)=y(1)=0 \tag{1.1}$$

with $\varphi(x)>0$. The eigenvalues of this problem form an unbounded sequence $\lambda_j(\varphi)>0$. If $f(x_1,x_2)$ is a real, differentiable function for $x_1,x_2>0$, define the functional $F(\varphi)$ by

$$F(\varphi)=f(\lambda_1(\varphi),\lambda_2(\varphi)). \tag{1.2}$$

Given measurable functions $a(x)$ and $b(x)$ such that $0<a(x)<b(x)$ on [0, 1], let

$$C(a(x),b(x))=\{\varphi(x)\in L_1[0,1]:a(x)\le\varphi(x)\le b(x)\}. \tag{1.3}$$

The extremal eigenvalue problem for $F(\varphi)$ can be formulated as follows: for what functions φ in $C(a(x),b(x))$ does $F(\varphi)$ achieve an extreme value? Willner and Mahar [1] and Gentry and Banks [2] derive general characterizations of those functions φ in $C(a(x),b(x))$ which extremize functions $f(\lambda_1,\cdots,\lambda_N)$. Willner and Mahar [3] and Keller [4] investigate the problem of extremizing the ratio of the first two eigenvalues when $a(x)$ and $b(x)$ are constant functions and give explicit characterizations of the extremizing functions $\varphi(x)$ as a function of the parameters a and b.

The extremal eigenvalue problem can be reformulated as a geometrical question. Given $a(x)$ and $b(x)$, characterize the set

$$\Lambda=\{(\lambda_1(\varphi),\lambda_2(\varphi)):\varphi\in C(a(x),b(x))\}. \tag{1.4}$$

As discussed in [1] and [2], Λ is compact. The extremal eigenvalue problem for arbitrary f is solved once Λ is known, in the sense that it is then reduced to the calculus problem of extremizing a function of two variables over a compact set.

In this paper we investigate the two-dimensional eigenvalue range Λ. After certain general results are derived for arbitrary $a(x)$ and $b(x)$, we specialize to the case with $a(x)$ and $b(x)$ constant to obtain additional results. The particular case $a(x)\equiv1$ and $b(x)\equiv4$ is then studied numerically. A proof that the set Λ corresponding to $C(1,4)$ has been determined is given under an assumption on the correctness of the numerical calculations. A characterization of the interior of Λ is also given. While the extremal eigenvalue problem for N eigenvalues, $N>2$, could be studied in a similar manner, we leave this problem to some future investigation.

The analysis is based on the following observation: if the function $f(x_1,x_2)$ does not achieve an extemum in the interior of Λ, the extremum must occur on the

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boundary. The condition that f have no interior extrema is

$$|\overline{\nabla} f|>0 \tag{1.5}$$

in the interior of Λ. As we are interested in determining the boundary of Λ, we will characterize those functions φ in $C(a(x),b(x))$ which extremize functions f satisfying (1.5).

2. Results for arbitrary bounding functions. For $\varphi(x)$ in $C(a(x),b(x))$, let λ_1 and λ_2 denote the first two eigenvalues of (1.1) with corresponding eigenfunctions $y_i(x)$ normalized by

$$\int_0^1 \varphi y_i^2 = 1, \qquad i=1,2.$$

Given f satisfying (1.5), define the function

$$g(\varphi,f,x) = -\lambda_1 \frac{\partial f}{\partial x_1} y_1^2(x) - \lambda_2 \frac{\partial f}{\partial x_2} y_2^2(x) \tag{2.1}$$

with the derivatives of f evaluated at (λ_1,λ_2). The results of [1], [2] and [4] include the following theorem.

THEOREM 1. *If $f(\lambda_1(\varphi),\lambda_2(\varphi))$ achieves its maximum over $C(a(x),b(x))$ at $\bar{\varphi}(x)$, then*:

i) $\bar{\varphi}(x)=b(x)$ *when* $g(\bar{\varphi},f,x)>0$,
ii) $\bar{\varphi}(x)=a(x)$ *when* $g(\bar{\varphi},f,x)<0$.

If f achieves its minimum over $C(a(x),b(x))$ at $\underline{\varphi}(x)$, then:

iii) $\underline{\varphi}(x)=a(x)$ *when* $g(\underline{\varphi},f,x)>0$;
iv) $\underline{\varphi}(x)=b(x)$ *when* $g(\underline{\varphi},f,x)<0$.

We will use Theorem 1 and Lemma 2.1, below, to characterize extremizing functions $\varphi(x)$.

LEMMA 2.1. *The function $g(\varphi,f,x)$ has either no zeros, one zero (simple or double), or two simple zeros for $x\in(0,1)$.*

Proof. If $\partial f/\partial x_1$ is zero at (λ_1,λ_2), then g has one double zero in $(0,1)$ since y_2 has one simple zero there . If $\partial f/\partial x_2$ is zero at (λ_1,λ_2), then g has no zeros on $(0,1)$. In the case that neither derivative of f vanishes, we rewrite (2.1) as

$$g(\varphi,f,x) = -\lambda_1 \frac{\partial f}{\partial x_1} y_1^2 \left[1 + \frac{\lambda_2}{\lambda_1} \cdot \frac{\partial f/\partial x_2}{\partial f/\partial x_1} \cdot r^2\right], \tag{2.2}$$

where $r(x) \equiv y_2(x)/y_1(x)$. We take $y_1(x)$ to be positive on $(0,1)$. The function $y_2(x)$ has one simple zero at $\bar{x}\in(0,1)$; we take $y_2>0$ on $(0,\bar{x})$. An elementary argument shows $r(x)$ is decreasing on $(0,1)$. Since $r(\bar{x})=0$, r^2 is decreasing on $(0,x)$ and increasing on $(\bar{x},1)$. Now (2.2) and the monotonicity properties of r^2 show that g can have at most one simple zero in $(0,\bar{x})$ and in $(\bar{x},1)$.

Combining Theorem 1 and Lemma 2.1, we obtain:

THEOREM 2. *The function $\varphi(x)$ which extremizes $f(\lambda_1,\lambda_2)$ has the following forms*:

$$\varphi(x)\equiv a(x) \quad \text{or} \quad \varphi(x)\equiv b(x), \qquad 0\le x\le 1 \tag{2.3a}$$

if $g(\varphi,f,x)$ never vanishes or has a double zero in $(0,1)$;

$$\varphi(x) = \begin{cases} a(x) & (b(x)), \qquad 0\le x<x_0, \\ b(x) & (a(x)), \qquad x_0<x\le 1 \end{cases} \tag{2.3b}$$

if g has a simple zero at $x_0 \in (0,1)$;

$$\varphi(x)=\begin{cases} a(x) & (b(x)), & 0\le x<x_0, \\ b(x) & (a(x)), & x_0<x<x_1, \\ a(x) & (b(x)), & x_1<x\le 1, \end{cases} \tag{2.3c}$$

if g has simple zeros at x_0, x_1 *in* $(0,1)$. *The sign of g at* $x=0$ *and the nature of the extremum determine whether* $\varphi(x)$ *starts as* $a(x)$ *or* $b(x)$.

We now use Theorem 2 to determine relationships between the eigenfunctions of (1.1) when φ extremizes some $f(\lambda_1,\lambda_2)$ over $C(a(x),b(x))$. There needn't be any special relationship in case (2.3a), as it is always possible to extremize some $f(\lambda_1,\lambda_2)$ by $\varphi(x)\equiv a(x)$ or $\varphi(x)\equiv b(x)$. This is called the no-jump case.

The possibilities in (2.3b) are called the one-jump (discontinuity) case. Since r^2 is decreasing on $(0,\bar{x})$ and increasing on $(\bar{x},1)$, it follows that

$$\left(\frac{y_2'(0)}{y_1'(0)}\right)^2 > \left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2 \ge \left(\frac{y_2'(1)}{y_1'(1)}\right)^2 \tag{2.4}$$

if $x_0<\bar{x}$ ($\bar{x}$ is the zero of y_2 in $(0,1)$), while

$$\left(\frac{y_2'(0)}{y_1'(0)}\right)^2 \le \left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2 < \left(\frac{y_2'(1)}{y_1'(1)}\right)^2 \tag{2.5}$$

if $\bar{x}<x_0$. Note that $\bar{x}=x_0$ cannot occur here. The possibilities in (2.3c) are called the two-jump case. We see from (2.2) and the definition of r that

$$\left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2 = \left(\frac{y_2(x_1)}{y_1(x_1)}\right)^2 \tag{2.6}$$

and $x_0<\bar{x}<x_1$ in this case.

3. Constant bounding functions. For the remainder of the paper, we assume $a(x)\equiv a^2$ and $b(x)\equiv b^2$ for some constants a and b. The constants appear as squares for convenience; furthermore, we will denote the first two eigenvalues by λ_1^2 and λ_2^2.

We use Theorem 2 in this special case to further characterize functions φ which extremize some $f(\lambda_1,\lambda_2)$ over $C(a^2,b^2)$. In the no-jump case, the functions $\varphi(x)\equiv a^2$ and $\varphi(x)\equiv b^2$ extremize certain functions f; nothing more can be said here.

If $\varphi(x)\in C(a^2,b^2)$, then $\varphi(1-x)\in C(a^2,b^2)$ and $\lambda_j(\varphi(1-x))=\lambda_j(\varphi(x))$ for all j. Thus, we may assume in the one-jump case that $\varphi(x)$ starts as b^2 and then switches to a^2. The jump point of $\varphi(x)$ is x_0, the location of the simple zero of g in $(0,1)$. We now rule out the possibility $\bar{x}<x_0$, $\bar{x}$ being the zero of y_2 in $(0,1)$, so the relations in (2.4) always hold in the one-jump case. With $\varphi(x)$ as above (i.e., b^2 and then a^2), define

$$I_j(x)=\left(y_j'(x)\right)^2+\lambda_j^2\varphi(x)y_j^2(x), \qquad j=1,2. \tag{3.1}$$

If we assume $\bar{x}<x_0$, then the relations in (2.5) hold. As the I_j are piecewise constant, we have

$$\frac{I_2(x_0^+)}{I_1(x_0^+)}=\left(\frac{y_2'(1)}{y_1'(1)}\right)^2 > \left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2. \tag{3.2}$$

However, using (2.4) and the identity

$$\lambda_2^2(a^2-b^2)y_2^2(x_0)=\left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2\lambda_2^2(a^2-b^2)y_1^2(x_0),$$

we easily verify that we also have

$$\frac{I_2(x_0^+)}{I_1(x_0^+)}<\left(\frac{y_2(x_0)}{y_1(x_0)}\right)^2. \tag{3.3}$$

Since (3.2) and (3.3) are inconsistent, we must have $x_0<\bar{x}$ and the relations in (2.4). Using piecewise trigonometric representations of the $y_j(x)$, the relations in (2.4) become

$$\left(\frac{\sin\lambda_2 bx_0}{\sin\lambda_1 bx_0}\right)^2<\left(\frac{\lambda_2}{\lambda_1}\right)\le\left(\frac{\sin\lambda_2 a(1-x_0)}{\sin\lambda_1 a(1-x_0)}\right)^2. \tag{3.4}$$

We now consider the two-jump case, so that

$$\varphi(x)=\begin{cases} c^2, & 0\le x<x_0,\\ d^2, & x_0<x<x_1,\\ c^2, & x_1<x\le 1,\end{cases} \tag{3.5}$$

where $c=a$ and $d=b$ or $c=b$ and $d=a$. Using piecewise trigonometric representations of the eigenfunctions $y_j(x)$, the extremization condition (2.6) can be rewritten as

$$\frac{1+\left(\frac{c}{d}\right)^2\cot^2\lambda_1 c(1-x_1)}{1+\left(\frac{c}{d}\right)^2\cot^2\lambda_2 c(1-x_1)}=\frac{1+\left(\frac{c}{d}\right)^2\cot^2\lambda_1 cx_0}{1+\left(\frac{c}{d}\right)^2\cot^2\lambda_2 cx_0}. \tag{3.6}$$

Thus, if $\varphi(x)$ given by (3.5) extremizes some f over $C(a^2,b^2)$, we have $x_0<\bar{x}<x_1$ and (3.6) (x_0 and x_1 are the zeros of g on (0.1)). It can be shown that $x_0=1-x_1$ when $c=a$ and $b=d$, so that the extremizing function $\varphi(x)$ is symmetric about $x=\frac{1}{2}$ in this case. As the proof is neither conceptually nor computationally enlightening, we omit the details. The proof consists of a lengthy sequence of elementary, albeit obscure, arguments. Details will be sent upon request.

We now summarize the results of this section. Assume the bounding functions $a(x)$ and $b(x)$ have the constant values a^2 and b^2. If $\varphi(x)\in C(a^2,b^2)$ extremizes some function $f(\lambda_1,\lambda_2)$ satisfying (1.5), one of the following possibilities must occur.

No-jump case.

$$\varphi(x)\equiv a^2 \quad\text{or}\quad \varphi(x)\equiv b^2. \tag{3.7}$$

One-jump case.

$$\varphi(x)=\begin{cases} b^2, & 0\le x<x_0,\\ a^2, & x_0<x\le 1,\end{cases} \tag{3.8}$$

$x_0<\bar{x}$, and

$$\frac{\sin\lambda_2 bx_0}{\sin\lambda_1 bx_0}<\frac{\lambda_2}{\lambda_1}\le\frac{\sin\lambda_2 a(1-x_0)}{\sin\lambda_1 a(1-x_0)}. \tag{3.9}$$

Two-jump case.

$$\varphi(x)=\begin{cases} a^2, & 0\le x<x_0,\\ b^2, & x_0<x<x_1,\\ a^2, & x_1<x\le 1,\end{cases} \tag{3.10a}$$

$x_0=1-x_1$, so $\varphi(x)$ is symmetric about $x=\frac{1}{2}$;

$$\varphi(x)=\begin{cases} b^2, & 0\le x<x_0,\\ a^2, & x_0<x<x_1,\\ b^2, & x_1<x\le 1,\end{cases} \tag{3.10b}$$

$x_0<\bar{x}<x_1$. Further,

$$\frac{1+\left(\frac{b}{a}\right)^2\cot^2\lambda_1 b(1-x_1)}{1+\left(\frac{b}{a}\right)^2\cot^2\lambda_2 b(1-x_1)}=\frac{1+\left(\frac{b}{a}\right)^2\cot^2\lambda_1 bx_0}{1+\left(\frac{b}{a}\right)^2\cot^2\lambda_2 bx_0}. \tag{3.11}$$

The numerical results of the next section show that extremizing functions $\varphi(x)$ need not be symmetric about $x=\frac{1}{2}$ in this subcase.

4. Numerical results. In this section we present the results of numerical computations for the particular example $a^2=1$ and $b^2=4$. If there exists (x_0,x_1) such that a function $\varphi(x)$ can be described by (3.10a), we call it type 1; if by (3.10b), type 2. By allowing the jump points to coincide either with each other or the boundary points 0 and 1, the no-jump and one-jump cases can be thought of as degenerate two-jump cases.

Figure 1 is a graph of the points in the x_0,x_1-plane which are the jump points of some extremizing $\varphi(x)$. In principle, a separate graph should be provided for the type 1 and type 2 cases. However, since $\varphi(x)$ and $\varphi(1-x)$ have the same eigevalues, we use the region $x_1\ge 1-x_0$ to graph type 1 points and $x_1\le 1-x_0$ to graph type 2 points. Since $x_1\ge x_0$, no points appear for $x_1<x_0$. The coordinates of all labeled points are given in Table 1 to four decimal places.

TABLE 1

	x_0	x_1	First eigenvalue	Second eigenvalue	Type
A	1.	1.	9.8696	39.4784	1
A	0.	1.	9.8696	39.4784	2
B	0.	1.	2.4674	9.8696	1
B	0.	0.	2.4674	9.8696	2
C	0.8465	1.	9.1063	28.9270	1
C	0.	0.8465	9.1063	28.9270	2
D	0.1174	0.8826	9.2305	30.0674	2
$P(S1)$	0.1471	1.	9.1967	29.6929	2
$P(S2)$	0.1195	0.8805	9.1967	29.6929	2
$P1$	0.1910	0.4270	4.6804	25.4354	1
$P2$	0.2550	0.8760	6.9479	20.6787	2
$P31$	0.0098	0.1469	9.1993	29.7190	1
$P32$	0.1464	0.9692	9.2019	29.7442	2
$P41$	0.0182	0.1469	9.2007	29.7434	1
$P42$	0.1464	0.9552	9.1926	29.6642	2

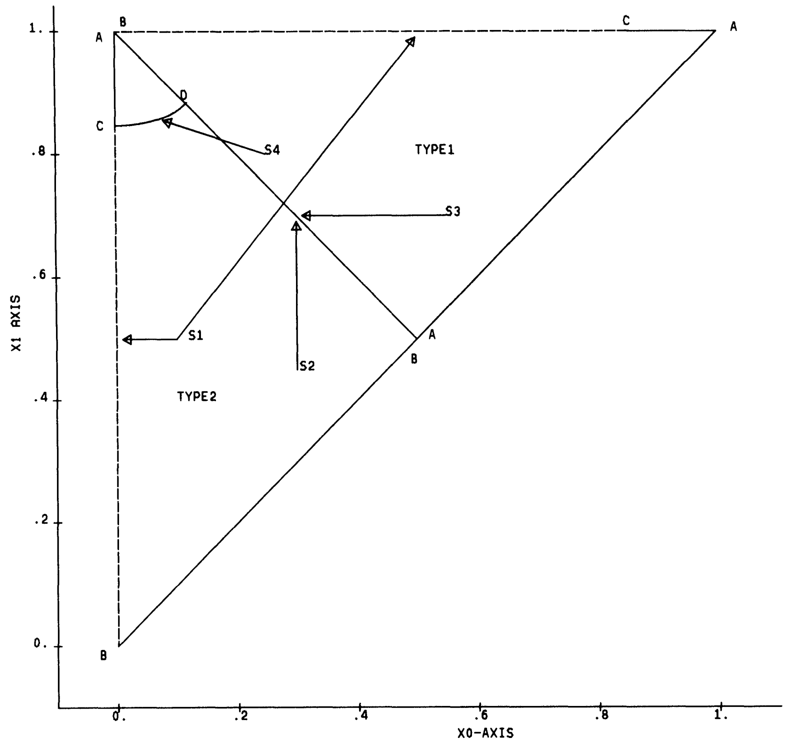

FIG. 1.

Type 1. In the nondegenerate two-jump case of type 1, we have $x_1 = 1 - x_0$. This line is labeled $S3$. The end points, (0, 1) and (.5, 5), are labeled B and A respectively. At B, $\varphi(x) \equiv b^2$, while $\varphi(x) \equiv a^2$ at A. The one-jump case is given by $x_0(1 - x_1) = 0$ and is labeled $S1$. The point (1, 1) is labeled A since $\varphi(x) \equiv a^2$ there. Any point on the line segment from (.5, .5) to (1, 1) is considered an "A" point and corresponds to the no-jump case. The one-jump case satisfies $x_0 < \bar{x}$ and (3.9). Only points between A and C along $S1$ satisfy these criteria. The dashed line from B to C indicates we already know these points do not correspond to possible extremizers.

Type 2. In this case, points where $x_0 = 1 - x_1$ are called B since $\varphi(x) \equiv b^2$. Similarly, (0, 1) is labeled A. All points on $\overline{BB}$ are in the no-jump case. The one-jump case is given by $\overline{BCA}$ as discussed previously, with no points on $\overline{CA}$ corresponding to extremizers.

We have numerically determined two curves in the (x_0, x_1)-plane which satisfy (3.11) in the two-jump case. These are the open line segment $\overline{AB}$ and the curve connecting C on the x_1-axis to D on the segment $\overline{AB}$. The point C divides the extremizing and nonextremizing one-jump cases. This second curve is labeled $S4$.

In Figs. 2–5 we present the points in the (λ_1, λ_2)-plane corresponding to φ's with jump points as in Fig. 1. Some points corresponding to other, nonextremizing, φ's are also shown. The coordinates of all labeled points are given in Table 1. Figure 2a shows the image of Fig. 1. The curves corresponding to $S1$–$S4$ are indicated, as well as the points A–D. Figure 2b is a pictographic representation of Fig 2a using only straight lines. The important features are highly exaggerated. A simple closed curve is formed by $S3$ from B to A, $S1$ from A to P, and $S2$ from P to B. We define R as the closed region bounded by this curve; subregions $R1$–$R4$ are defined as shown in Fig. 2c. Figures 3, 4 and 5 are blow-ups of the subrectangle from Fig. 2.

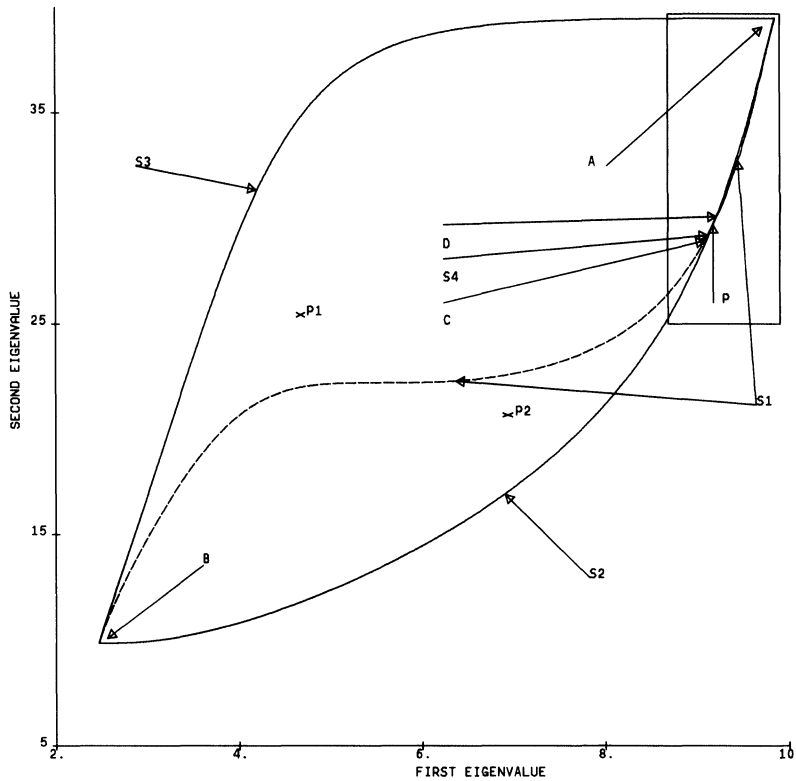

FIG. 2A.

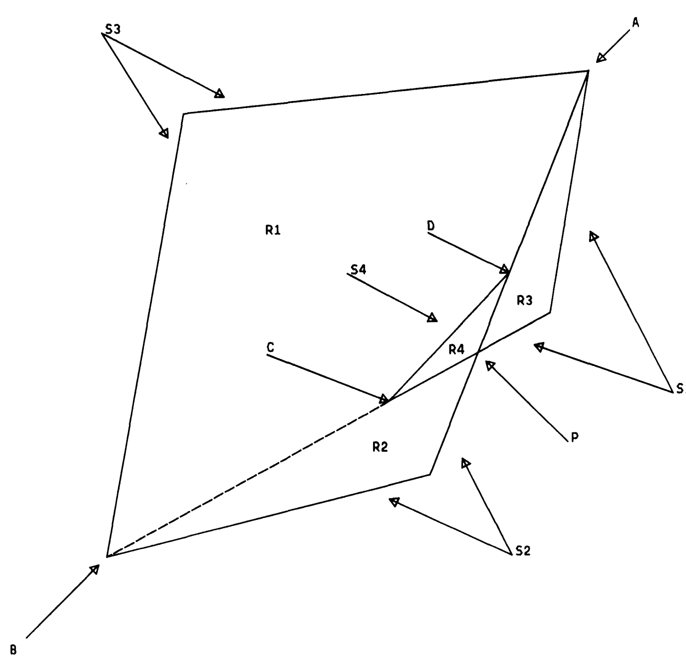

FIG. 2B. *Pictographic representation.*

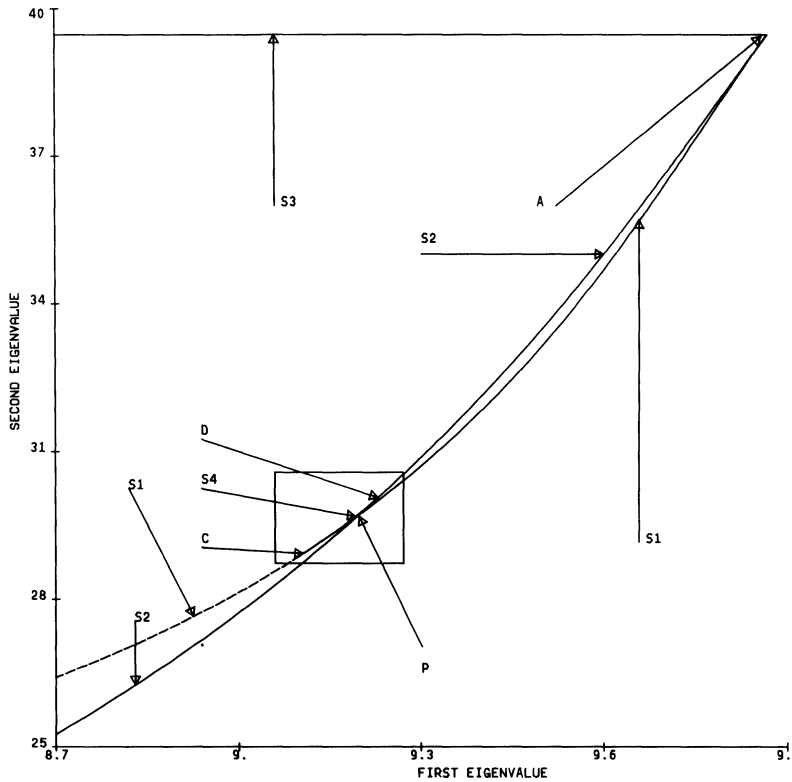
40
37
34
31
28
25
SECOND EIGENVALUE
8.7
9.
9.3
9.6
9.9
FIRST EIGENVALUE
S3
A
S2
S1
D
S1
S4
C
S2
P

FIG. 3.

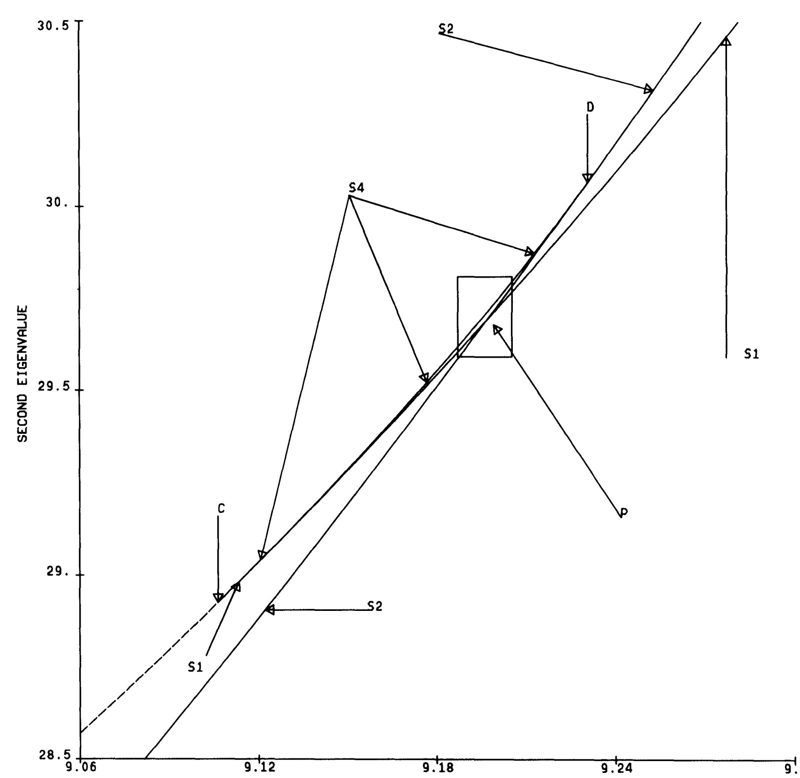
30.5
30.
29.5
29.
28.5
SECOND EIGENVALUE
9.06
9.12
9.18
9.24
9.3
FIRST EIGENVALUE
S2
D
S4
S1
P
C
S2
S1

FIG. 4.

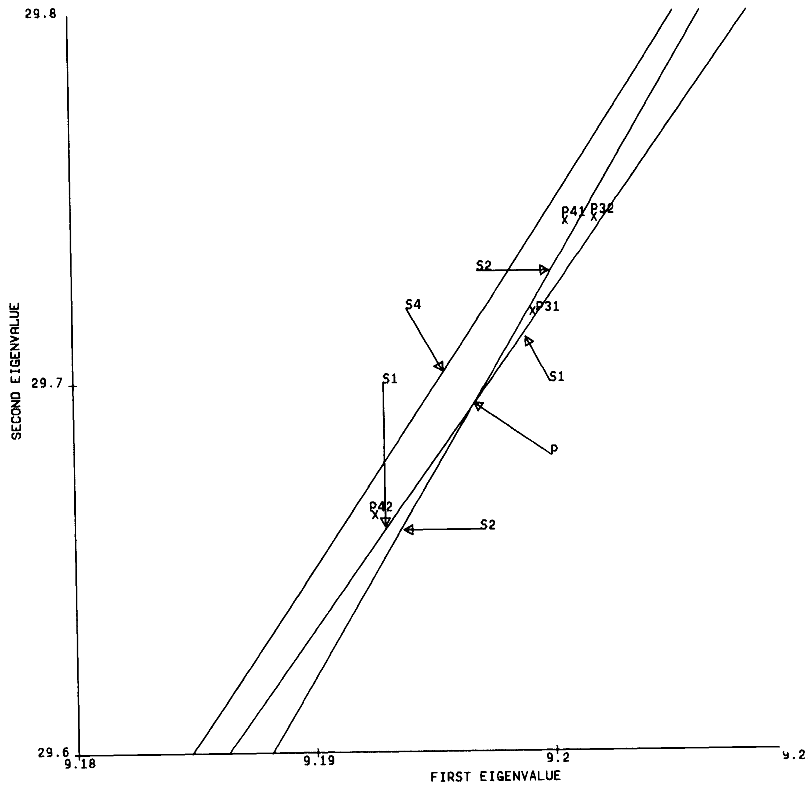

FIG. 5.

5. Λ for $C(1,4)$. We now prove that the region R in Fig. 2a is Λ, the two-dimensional eigenvalue range, for $C(1,4)$. Assuming that the numerical calculations are sufficiently accurate, Newton's method could be used to prove the existence of eigenvalues close to the computed ones. Numerical experience suggests that the computational accuracy can be substantially increased, so we assume the numerical results are correct.

THEOREM 3. *The region R is the two-dimensional eigenvalue range Λ for $C(1,4)$. That is, $(\lambda_1,\lambda_2)\in R$ if and only if there exists $\varphi\in C(1,4)$ such that* (1.1) *has λ_1 and λ_2 as its first two eigenvalues.*

Proof. First, assume (λ_1,λ_2) belongs to Λ but not R, and let $(\tilde\lambda_1,\tilde\lambda_2)$ be a point exterior to Λ and R such that it can be connected to (λ_1,λ_2) by a curve which does not intersect R. There exists $(\bar\lambda_1,\bar\lambda_2)$ on this curve which is a boundary point of Λ. Let $(\sigma_n,\tau_n)\in\Lambda$ converge to $(\bar\lambda_1,\bar\lambda_2)$ and set $f_n(\lambda_1,\lambda_2)=(\sigma_n-\lambda_1)^2+(\tau_n-\lambda_2)^2$. The points Q_n in Λ which minimize the f_n must lie on one of the S-labeled curves. However, since $(\bar\lambda_1,\bar\lambda_2)$ is outside R, there exists $\delta>0$ such that $\underline{f_n}$ evaluated at Q_n is no smaller than δ for all n. This is impossible since $(\sigma_n,\tau_n)\to(\bar\lambda_1,\bar\lambda_2)\in R$. Thus, if $(\lambda_1,\lambda_2)\in\Lambda$, then $(\lambda_1,\lambda_2)\in R$ also.

Now, let $\mathfrak{X}=(\lambda_1,\lambda_2)\in R$. If $\mathfrak{X}$ lies on some S curve, it is in Λ by definition of the S curves. If $\mathfrak{X}$ is not on any of these curves, it must be in one of the four subregions of R. As all cases are similar, assume $\mathfrak{X}\in R1$. Connect $\mathfrak{X}$ and $P1$ in $R1$ with an arc lying entirely in $R1$. If $\mathfrak{X}\notin\Lambda$, there exists $\bar{\mathfrak{X}}$ on this arc which is a boundary point of Λ. Let $\mathfrak{X}_n\notin\Lambda$ converge to $\bar{\mathfrak{X}}, \mathfrak{X}_n=(\sigma_n,\tau_n)$. Again consider minimizing the functions f_n. The minimizing points Q_n all lie on S curves. Since $\bar{\mathfrak{X}}$ is in the interior of $R1$, the $\mathfrak{X}_n$ are also for n large. Thus, there exists $\delta>0$ such that the f_n evaluated at Q_n are no smaller than δ. But, $\mathfrak{X}_n \to \bar{\mathfrak{X}} \in \Lambda$, and so the f_n at Q_n must approach zero. Thus, $\mathfrak{X} \in R$ implies $\mathfrak{X} \in \Lambda$.

Theorem 3 shows that R and Λ are the same, so we have determined the two-dimensional eigenvalue range for $C(1,4)$.

THEOREM 4. *Every point in the interior of* $R1$ *corresponds to a* φ *of type* 1, *but not type* 2. *Every point in the interior of* $R2$ *corresponds to a* φ *of type* 2, *but not type* 1. *Points in the interior of* $R3$ *and* $R4$ *correspond to a* φ *of type* 1 *and a* φ *of type* 2.

Proof. We define $PC(1,4)$ as the class of functions $\varphi(x)$ described in (3.5), with $c=1$ and $d=2$ or $c=2$ and $d=1$, such that $0<x_0<x_1<1$. Elements of $PC(1,4)$ have two jumps. By utilizing piecewise trigonometric representations of the corresponding eigenfunctions $y_i(x)$ for (1.1), tedious algebraic and trigonometric manipulations can be used to show that

$$\left|\frac{\partial(\lambda_1,\lambda_2)}{\partial(x_0,x_1)}\right| \neq 0 \tag{5.1}$$

if the extremization condition (3.11) is not satisfied. Here, x_0 and x_1 are the jump points of φ and λ_1, λ_2 are the first two eigenvalues of the corresponding problem (1.1). Condition (5.1) guarantees that the implicit function theorem can be applied.

Let $\Omega 1$ be the eigenvalue image of type 1 elements of $PC(1,4)$, so $c=1$ and $d=2$. Assume (λ_1,λ_2) is not in $\Omega 1$, but is interior to $R1$. Connect (λ_1,λ_2) to $P1 \in \Omega 1$ with an arc interior to $R1$. There exists a point $\mathfrak{X}$ on this arc which is a boundary point of $\Omega 1$. Since $\mathfrak{X} \in R1$, the implicit function theorem implies $\mathfrak{X} \notin \Omega 1$. Let $\mathfrak{X}_n \in \Omega 1$ converge to $\mathfrak{X}$. The points $\mathfrak{X}_n$ are generated by φ_n in $PC(1,4)$ of type 1 with jump points $(x_{0,n}, x_{1,n})$. Choosing subsequences if necessary, let $x_{0,n} \to x_0$ and $x_{1,n} \to x_1$. Use (x_0,x_1) to construct $\varphi(x)$ in $PC(1,4)$ of type 1 such $\varphi_n \to \varphi$ in the L_1-norm. Now, $\mathfrak{X}=(\lambda_1(\varphi),\lambda_2(\varphi))$ by continuity of the eigenvalues [5]. If $0<x_0<x_1<1$ and (3.11) is not satisfied, we have the contradiction $\mathfrak{X} \in \Omega 1$. If (3.11) is satisfied, or φ is a degenerate two-jump function, $\mathfrak{X}$ would be on an S curve, another contradiction. Thus, $\mathfrak{X}$ is not a boundary point of $\Omega 1$, the final contradiction. Hence, (λ_1,λ_2) in the interior of $R1$ implies (λ_1,λ_2) in $\Omega 1$. By similar reasoning, every point in the interior of $R2$ is the image of a type 2φ, while points in the interior of $R3$ and $R4$ are images of a φ of type 1 and a φ of type 2.

We now show that no point in the interior of $R2$ is the image of a type 1 φ. Assume some type 1 φ has its eigenvalues in the interior of $R2$. As above, we then have that every element of the interior of $R2$ is the image of a type 1 φ. Let $\mathfrak{X}=(\lambda_1,\lambda_2)$ be a point on $S2$ in the interior of the segment BP, and let $\mathfrak{X}_n$ in $\Omega 1$ converge to $\mathfrak{X}$. The points $\mathfrak{X}_n$ are generated by φ_n in $PC(1,4)$ of type 1 with jump points $(x_{0,n}, x_{1,n})$. Choosing subsequences, $x_{0,n} \to x_0$ and $x_{1,n} \to x_1$, we use (x_0,x_1) to construct φ in $PC(1,4)$ such that $\varphi_n \to \varphi$ in the L_1-norm. Thus, $\mathfrak{X}=(\lambda_1(\varphi),\lambda_2(\varphi))$ by the continuity of the eigenvalues [5]. If $x_0=1-x_1$, $\mathfrak{X}$ is on $S3$, a contradiction. If φ is a degenerate two-jump function, $\mathfrak{X}$ lies on AB along $S1$, another contradiction. Thus, (5.1) holds and the implicit function theorem shows there is a neighborhood about $\mathfrak{X}$ corresponding to a neighborhood about (x_0,x_1). This is false since no point of $\Omega 1$ can lie outside $\Lambda = R$. Similarly, no point of $R1$ corresponds to a φ of type 2.

6. Conclusions. This paper has characterized the set of points $(\lambda_1(\varphi),\lambda_2(\varphi))$ as $\varphi(x)$ varies over $C(a(x),b(x))$. It seems curious that this question hasn't been studied already, since it is of theoretical interest to know precisely what the various eigenvalue ranges are. Although a complete characterization has been given only for the case

$a(x)\equiv 1$ and $b(x)\equiv 4$, the general results suggest similar characterizations hold for arbitrary choices of constant bounding functions. The case of nonconstant bounding functions does not appear to be amenable to the same sort of detailed analysis.

The results presented above also provide a method, simple in principle and not difficult numerically, for solving extremal eigenvalue problems with constant bounding functions. Compute all pairs $(\lambda_1(\varphi), \lambda_2(\varphi))$ with φ one of the possible extremizers given in (3.7)–(3.11). A set of curves SR will be generated. The answer to the extremal eigenvalue problem will lie on one of these curves if (1.5) is satisfied. If (1.5) is violated, a piecewise constant φ with two jumps such that (3.11) is violated, or a one-jump φ such that $\bar{x}\leq x_0$, will provide the answer. In all cases, the extremizing φ will be piecewise constant, have at most two jumps, and assume only the smallest and largest allowed values.

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